Krishnaa Vadivel | Scientific Computing | Machine Learning | Electrical Engineering

516-853-5663 • srikrishnaa.careers@gmail.com • sri-krishnaa-ece-phd
Website | GitHub: krishnaa423 | LeetCode: krishnaa42342 | Docker Hub: krishnaa42342

Profile

Broad-profile electrical engineering PhD researcher with training and project work spanning computational physics, semiconductor physics, integrated photonics, scientific machine learning, embedded systems, digital design, and distributed scientific software. The core strength is developing theoretical models and implementing software across research and engineering settings.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher *2021–present*

- Developed machine-learning and automatic-differentiation models in Python for material-property prediction and optical-emission calculations.
- Developed first-principles and quantum many-body simulation software in Python and C++ using MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm, and CMake for large simulation and linear-algebra workloads.
- Built distributed and accelerator-enabled applications for Frontier, Perlmutter, Frontera, and related clusters, spanning both NVIDIA and AMD GPU systems.
- Worked on light-matter interaction and quantum many-body modeling, including self-trapped-exciton and exciton-polaron problems relevant to semiconductor and photonic materials.
- Helped fabricate and characterize semiconductor materials during the early PhD, alongside simulation and modeling work tied to photonic and electronic materials.
- Built agentic AI pipelines for reading papers and software documentation, then generating structured inputs for first-principles simulation workflows.
- Support embedded-systems instruction and hardware-oriented lab work through teaching-assistant responsibilities tied to firmware, debugging, and instrumentation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS talks in 2024, 2025, and 2026.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer *2019–2021*

- Built parallel applications with CUDA and OpenCL fallback for large acoustic waveform and sensor datasets used in engineering diagnostics and field-tool analysis.
- Developed embedded software and digital logic for sensing, logging, and control functions using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-based systems.
- Supported board- and system-level engineering tasks including oscilloscope-based debugging, integration testing, field validation, and selective PCB modifications in Altium.
- Developed full-stack server applications with ASP.NET and Microsoft SQL Server for real-time control of company tool testing, data collection, and data processing.
- Developed C# GUI applications for preparing customer inputs and interacting with deployed hardware systems.
- Used `dolfinx`-based finite-element analysis for frequency analysis of steel pipes in wells.

FindLight

Photonics Technical Writer *2018–2019*

- Produced technical articles and engineering-facing content covering laser bench systems, optics, photonics components, and optoelectronic systems.

Selected Projects

Machine Learning Models for Material Properties

- Developed PyTorch Geometric graph neural networks and automatic-differentiation models to accelerate material-property and optical-emission calculations in molecular and condensed-matter systems.

Semiconductor Device Simulation

- Built numerical treatments of electrostatics and transport equations, including Poisson, drift-diffusion, and pn-junction-style models linking semiconductor theory with simulation practice.

Integrated Photonics Bandpass Filter

- Developed Meep FDTD simulations for a ring-resonator-based integrated-photonics bandpass filter that was later fabricated and tested with an erbium-doped fiber laser.

Container Orchestration for First-Principles Computing

- Developed CUDA-ready and cross-environment containers for first-principles software stacks and published them to Docker Hub for portable scientific computing.

Embedded Robotics and Human-Interface Projects

- Built robotics and embedded-system class projects including a surface-EMG robotic hand that was later published in Circuit Cellar, plus maze-navigation work with wireless coordination.

Integrated Photonics and FDTD Modeling

- Simulated band-pass and resonator-style photonic structures with Meep-based FDTD workflows for integrated-optics exploration.

Agentic Scientific Automation

- Built agentic AI pipelines that read software documentation and research papers to generate structured inputs and workflow scaffolds for computational science.

Education

Yale University
PhD, Electrical Engineering *2021–present*
Ongoing work spanning computational materials, many-body theory, semiconductor-relevant modeling, and scientific software.

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*
Included stochastic-computing-related work, device-oriented EE topics, and advanced design coursework.

Cornell University
BS, Electrical and Computer Engineering *2013–2017*
Included embedded systems, robotics, controls, and strong math/physics/computing preparation.

Selected Publications

2025: de Paula, A. M., Li, S., Hou, B., Vadivel, S., Teles-Ferreira, D. C., Iudica, A., Kabacinski, P., Hosseini, H., McArthur, J., Cerullo, G., et al. (2025). Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites. *Journal of the American Chemical Society, 147*(32), 28923–28931.

Selected Conference Presentations

2026: Vadivel, S., Grande, R. D., Strubbe, D. A., & Qiu, D. Y. (2026). First-principles study of exciton-polarons and self-trapped excitons. *APS Global Physics Summit 2026*.

2025: Vadivel, S., Qiu, D. Y., Grande, R. D., & Strubbe, D. A. (2025). First-principles study of self-trapped exciton formation in double perovskites. *APS Global Physics Summit 2025*.

2024: Vadivel, S., & Qiu, D. (2024). First-principles study of self-trapped exciton formation in double perovskites using excited state forces. *APS March Meeting Abstracts 2024*, F01–007.

Selected Coursework

Mathematics: Differential Geometry; Abstract Algebra; Honors Mathematical Analysis II

Computer Science: Theoretical Computer Science; Probabilistic Machine Learning

Physics: Graduate Quantum Mechanics I–II; Solid State Physics I–II; Many-Body Quantum Physics

Electrical Engineering: Semiconductor Physics; Thin Film Science; Lasers and Optoelectronics; Integrated Photonics; VLSI Design; Analog VLSI Design; Mathematics of Signals and Systems; Advanced Embedded Systems

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, graph neural networks, transformers, automatic differentiation, agentic AI

Hardware: Verilog, Lattice FPGA, PIC32, dsPIC, STM32, C8051, sensor interfaces, PCB bring-up, oscilloscope debugging

Software / Systems: ASP.NET, Microsoft SQL Server, Linux command line, Docker, Docker Compose, Kubernetes, LangChain, OpenAI API, Ollama, gcloud

HPC Platforms: Frontier, Frontera, Perlmutter, and related national-lab or university HPC environments